



ChatGPT for chatting and searching: Repurposing search behavior

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ABSTRACT

Generative AI tools, exemplified by ChatGPT, are transforming the way users interact with information by enabling dialogue-based querying instead of traditional keyword searches. While this conversational approach can simplify user interactions, it also presents challenges in structuring effective searches, refining prompts, and verifying AI-generated content. This study addresses these complexities by repurposing traditional search tactics for use in conversational AI environments, specifically to support the Searching as Learning (SaL) paradigm. Forty-five adapted tactics are introduced to aid users in defining information needs, refining queries, and evaluating ChatGPT's responses for relevance, utility, and reliability. Using the Efficient Search Tactic Identification (ESTI) method and constant comparison analysis, these tactics were mapped into a stratified model with seven categories. The framework provides a structured approach for users to leverage conversational agents more effectively, promoting critical thinking and iterative learning. This research underscores the importance of developing robust search strategies tailored to conversational AI environments, facilitating deeper learning and reflective information engagement. Additionally, it highlights the need for ongoing research into the design and evaluation of future chat-and-search systems.

1. Introduction

With the rise of generative Artificial Intelligence (AI) tools like ChatGPT, online information retrieval may trend away from traditional keyword-based searches toward more conversational, dialogue-driven approaches. This transition would offer a more intuitive way for users to query information, bypassing the need for specialized syntax or rigid keyword formatting. Rather than shaping queries to fit the limitations of search engine algorithms, users could engage in natural language conversations that encourage iterative dialogue and facilitate more personalized responses (Brown, 2020; Silver et al., 2016). However, despite this convenience, the conversational nature of large language models (LLMs) presents new challenges in structuring effective searches and validating the reliability of AI-generated responses. Traditional search engines typically require users to refine queries using Boolean operators or filter mechanisms (Hearst, 2009, p.1–3), often with multiple search iterations to locate relevant results (Hearst, 2009, p.7–11). This structured approach to information-seeking contrasts with the open-ended, dynamic interactions possible with conversational AI (Kasneji et al., 2023), where users must now focus on prompting strategies that influence the AI's response quality and relevance. The shift

introduces complexities, as users must adjust to a new search paradigm where question phrasing, prompt sequencing, and response verification become central elements of effective information retrieval.

In this evolving landscape, search tactics that users once relied on, such as using Boolean logic operators to broaden or narrow search parameters (Thatcher, 2006) or selecting specific terms to refine searches (Bates, 1979b), require adaptation to the unique demands of tools powered by LLMs. A critical evaluation of how these traditional tactics could be repurposed for conversational AI environments is important to ensure that users can navigate information effectively, maintain control over search direction, and foster deeper understanding in dialogue-driven interactions. Existing approaches like “Searching as Learning” highlight the value of iterative searches for building comprehensive understanding (Rieh et al., 2016; Vakkari, 2016), but generative AI tools must go further, supporting not only search success but also critical thinking and knowledge synthesis (Marchionini, 2006).

To address this gap, the present study identified and adapted 45 traditional search tactics to facilitate more effective user interactions in generative AI environments. These repurposed tactics aimed to help users convey their information needs and search objectives in more meaningful ways, enabling conversational agents to respond with

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contextualized, relevant information. The study also drew attention to “Searching as Learning” strategies, assessing how they can be integrated into ChatGPT-like tools to promote deeper understanding, critical assessment, and iterative refinement of information.

2. Problem statement

The transition back to a more natural form of querying, asking questions as humans are inclined to do, promises to redefine the interaction with information technology, as highlighted by Croft et al. (2010) and Ram (1991) in seminal studies. Despite the intuitive appeal of this shift, the integration of LLM-powered AIs as search partners introduces new complexities and challenges that necessitate a thorough examination.

The core of this study lies in understanding these complexities, particularly how they altered the dynamics of the search process. Traditional search engines have conditioned users to distill their queries into keywords, a practice shaped more by technical limitations than by human inclination (Croft et al., 2010, p.75). In contrast, conversational AI interfaces invite users to engage in a dialogue, potentially reducing the need to browse multiple sources for information. This change, while enhancing efficiency, raises concerns about the depth of knowledge acquisition and the potential for an uncritical acceptance of synthesized responses.

Additionally, this transformation in search behavior brings to the forefront the necessity of reevaluating and repurposing search tactics and strategies. The study was guided by the following research questions:

RQ1. How can search tactics and strategies be repurposed to perform a prompt process with the aid of a tool such as ChatGPT?

RQ2. What specific search tactics and strategies could be repurposed for this type of prompting?

Addressing these questions necessitated a comprehensive exploration of the intersection between conversational AI and information retrieval practices. The study aimed to achieve this goal by conducting an extensive literature review and applying it within a novel user-system interaction framework.

3. Literature review

3.1. Theoretical framework

ChatGPT, as Chomsky et al. (2023) pointed out, is “unable to balance creativity with constraint”, either overgenerating (e.g., retrieving both truths and falsehoods) or undergenerating (e.g., exhibiting noncommitment behavior) information. How can a user navigate such an environment? There are three aspects that are pertinent to search systems and should be considered when using LLM-based systems as an information-seeking tool. The first is the anomalous state of knowledge (ASK) (Belkin et al., 1982), in which users’ information needs arise from the lack of their knowledge. Belkin et al. (1982) referred to ASK as an “anomaly in the user’s state of knowledge concerning some topic or situation”, hence the acronym. In this particular state, users have trouble expressing or structuring their informational problem. Vakkari (2010) pointed out that this ill-structured informational problem caused by ASK starts with the lack of information needed to define the problem and devise a solution to it. ASK also has no single right approach for its solution, suffers the impact of new gathered information that changes the problem definition, and since it has no correct solution, it is a gradual learning process to devise one (Vakkari, 2010).

In this sense, it is part of the “job” of a conversational agent to elicit users’ actual need, to aid them to track the impact of the gathered information on the informational problem definition, and to assist in devising a solution by helping them formulate the prompts more clearly

during their interactions. Second, in a chat-and-search process, the conversational agent must identify changes in the user’s knowledge (Tibau et al., 2023). This can be achieved either by analyzing the user’s behavior or by eliciting information about how the user utilizes the sources they encounter to shape their understanding. Multi-turn conversational interactions are particularly well-suited for this purpose, as the progression of the conversation provides a clear structure for tracking and evaluating knowledge development.

Third, in searches suited for multi-turn conversational interactions, the search results must guide the responses of the conversational agent. Radlinski and Craswell (2017) suggested that to satisfy a user’s information need, AI must interact with a set of results to produce a single-item response. This third aspect demands that the conversational agent be able to evaluate the need to retrieve the control of the conversation from users or pass it to them depending on its set state. The control of the conversation usually passes from one side to the other through assertions, commands, questions, and prompts (Radlinski & Craswell, 2017). Still, the decision to do it depends on the user’s ability to evaluate the prompts and the information they provide, and the AI’s ability to select the proper subsets of the result set.

3.2. Search behaviors, tactics, and strategies

Searching for information online involves distinct components: behaviors, tactics, and strategies. Search behaviors refer to the actions users take during the search process, as well as their thought processes while searching (Bates, 1979b). Search tactics, a subset of these behaviors, are defined as specific moves a user makes to further their search (Smith, 2012), characterized as a set of temporally and semantically related search actions (Wildemuth, 2004). In essence, search tactics represent every deliberate step taken toward locating the desired information (Bates, 1979b).

Search strategy, on the other hand, refers to the planning and organization of a search, which may occur either in advance or during the search itself (Smith, 2012). In addition, the study of search strategies encompasses the theoretical and practical principles behind constructing and applying search strategies and tactics (Bates, 1979b).

Online search strategies can be divided into three main domains (Reisoğlu et al., 2019). The behavioral domain involves basic skills for interacting with and navigating the Web; the procedural domain focuses on general search techniques and problem-solving during the search process; and the metacognitive domain deals with monitoring, identifying key information, and evaluating the information retrieved (Reisoğlu et al., 2019).

Search tactics are typically grouped into four main categories. Monitoring tactics help keep the search efficiently on track (Bates, 1979b); file structure tactics involve techniques for using database structures to find information (Smith, 2012); search formulation tactics assist in designing or refining search commands (Bates, 1979b); and term tactics guide the selection and revision of specific terms used in the search formulation (Bates, 1979b; Smith, 2012).

3.3. Related work

In recent years, researchers have examined the impact of ChatGPT and other generative AI tools across various fields, such as academic publishing, clinical decision support, and code debugging. For instance, studies have explored ChatGPT’s role in academic integrity, with a focus on its performance in online exams (Susnjak, 2022), and its effectiveness in medical settings, such as generating discharge summaries (Patel & Lam, 2023), and supporting clinical decisions (Sandmann et al., 2024). Additionally, there has been research into ChatGPT’s capabilities for literature reviews (McGee, 2023; Wang et al., 2023) and bug fixing in code (Surameery & Shakor, 2023).

Despite these advancements, one area remains underexplored: the impact of generative AI on Web searching strategies. While some

research has investigated search term suggestions (Sutton & Parisi, 2024), search engine optimization (Cutler, 2023), and search engine reliability (Memon & West, 2024), the broader implications for Web searching behaviors and strategies have received comparatively little attention. This study sought to address that gap by focusing on how ChatGPT and similar tools reshape search strategies, moving away from traditional query-based searches to more dynamic, conversational interfaces.

Web searching remains a fundamental method for information seeking and learning. This study adopted the “Searching as Learning” paradigm, which emphasizes the iterative, exploratory nature of searches as users engage in content exploration and decision-making processes (Rieh et al., 2016; Vakkari, 2016). By understanding how users’ cognitive efforts interact with search systems, this paradigm highlights the evolving relationship between users and information systems (Tibau et al., 2023), particularly with the introduction of generative AI models. The repurposing of traditional search tactics for this new environment is crucial for maintaining effective information retrieval practices in the age of conversational AI.

4. Methodology

4.1. Research design

In this study, the researchers selected constant comparison analysis as the primary qualitative data analysis method for its ability to capture a spectrum of interpretations and systematically identify relationships within the data. This iterative method was particularly valuable for examining new behaviors associated with LLM tools, as it allowed continuous comparison of emerging insights against previously categorized elements. When used alongside Saracevic’s (1997) stratified model, constant comparison analysis enabled the researchers to ground these insights in established search tactics, ensuring that novel LLM-specific behaviors were contextualized within traditional frameworks. This approach highlighted the convergence of new and established strategies and helped uncover potential gaps in current search tactics, reinforcing the study’s theoretical and practical contributions (Strauss & Corbin, 1998).

The research process followed a systematic path, beginning with a comprehensive review of relevant literature to identify existing search tactics and strategies. Onwuegbuzie et al.’s (2012) method for qualitative data analysis guided this review, which emphasized careful categorization of identified strategies and systematic comparisons to reveal new insights.

In adapting classic search tactics for ChatGPT’s prompt-response cycle, the researchers considered both behavioral and procedural domains, which typically involve term-based searches, as well as the metacognitive domain. The researchers repurposed behavioral and procedural tactics, such as search formulation and monitoring, to align with ChatGPT’s conversational interface, which emphasized text-based responses over hyperlink navigation (Dwivedi et al., 2023).

The researchers repurposed the metacognitive domain to address ChatGPT’s well-known limitation: its tendency to present information, whether accurate or not, with equal confidence (Dwivedi et al., 2023). Tactics in this domain focused on evaluating the relevance, usefulness, and trustworthiness of AI-generated content. In addition, the researchers incorporated indicators to further organize the strategies and provide insight into how effectively users can access and evaluate information sources (Reisoglu et al., 2019).

4.2. The mapping process

To choose and group the tactics, the researchers considered the differences between the systems’ interfaces. Also, the researchers took steps to avoid interpretation bias, since the interpretation of the tactic’s meaning and use in search engines and its subsequent selection to be

repurposed to LLM tools may have lead to conclusions out of context. Search tactics and strategies could be abstracted away from interface interference, but doing so would leave the system’s peculiarities out of the picture and its influence in shaping the user’s behavior would not be considered (Thatcher & Milo, 2014). In this sense, the researchers needed a comparative method.

The researchers used the ESTI method to map tactics (He et al., 2016), which was designed to provide interpretability and comparability of user actions across different search systems. In combination with Saracevic’s (1997) layered perspective and the constant comparison approach, ESTI ensured that tactics were classified based not just on action, but on purpose and context within LLM environments in three distinct mapping situations:

- **One-to-one mapping:** A single user action directly corresponded to one specific tactic. This type of mapping was straightforward and typically occurred in simple user-system interactions. For example, when a user entered a specific search term in a prompt on ChatGPT, this action directly corresponded to the “Search Formulation” tactic. The one-to-one mapping ensured that the user’s input action immediately affected the system’s response.
- **Many-to-one mapping:** A single tactic can be applied through various user actions. For instance, a user might explore multiple pathways to check for inconsistencies in ChatGPT’s response, such as rephrasing the question or requesting additional sources. Although these actions differ, they all corresponded to the “Crosscheck” tactic, which was aimed at verifying the accuracy of the information provided.
- **One-to-many mapping:** A single user action can correspond to multiple tactics, depending on the context of the search. For example, the same action, such as hovering over a search result, could represent different tactics depending on whether the user was skimming or closely analyzing the content. Similarly, when a user revised a prompt to add more detail or context, the action could be classified under both the “Refine” and “Specialization” tactics, depending on whether the goal was to clarify the response or narrow the search focus.

4.3. Data collection methods

The data were collected through a literature review of papers focusing on Library and Information Science (LIS) across three repositories: Digital Commons Network, [arXiv.org](https://arxiv.org), and Google Scholar. The first two prioritized open access, significantly facilitating the acquisition of papers without the need for subscription barriers. Google Scholar indexed a wide range of sources from various disciplines, which ensured access to a broad spectrum of academic work. Additionally, papers recommended during the peer review process to complement the original literature search were included in the “Reviewer Recommendations category”. The review focused on works that examined the evolution of search strategies in the context of digital library use and Web searching, the cognitive processes involved in information seeking, and how advancements in technology have transformed user search behaviors. Fig. 1 provides a comprehensive summary of the literature review process.

4.4. Data selection methods

The researchers selected the papers assessed for the literature review based on inclusion criteria to cover the development of search tactics, strategies, and their operationalization, while also considering the impact of technology, user cognition, and various evaluation methodologies. Appendix A provides the inclusion criteria and their description. In this phase, the researchers examined the titles and abstracts of the papers. Where necessary, the researchers also reviewed the introduction section to gain a clearer comprehension of the study. The papers selected

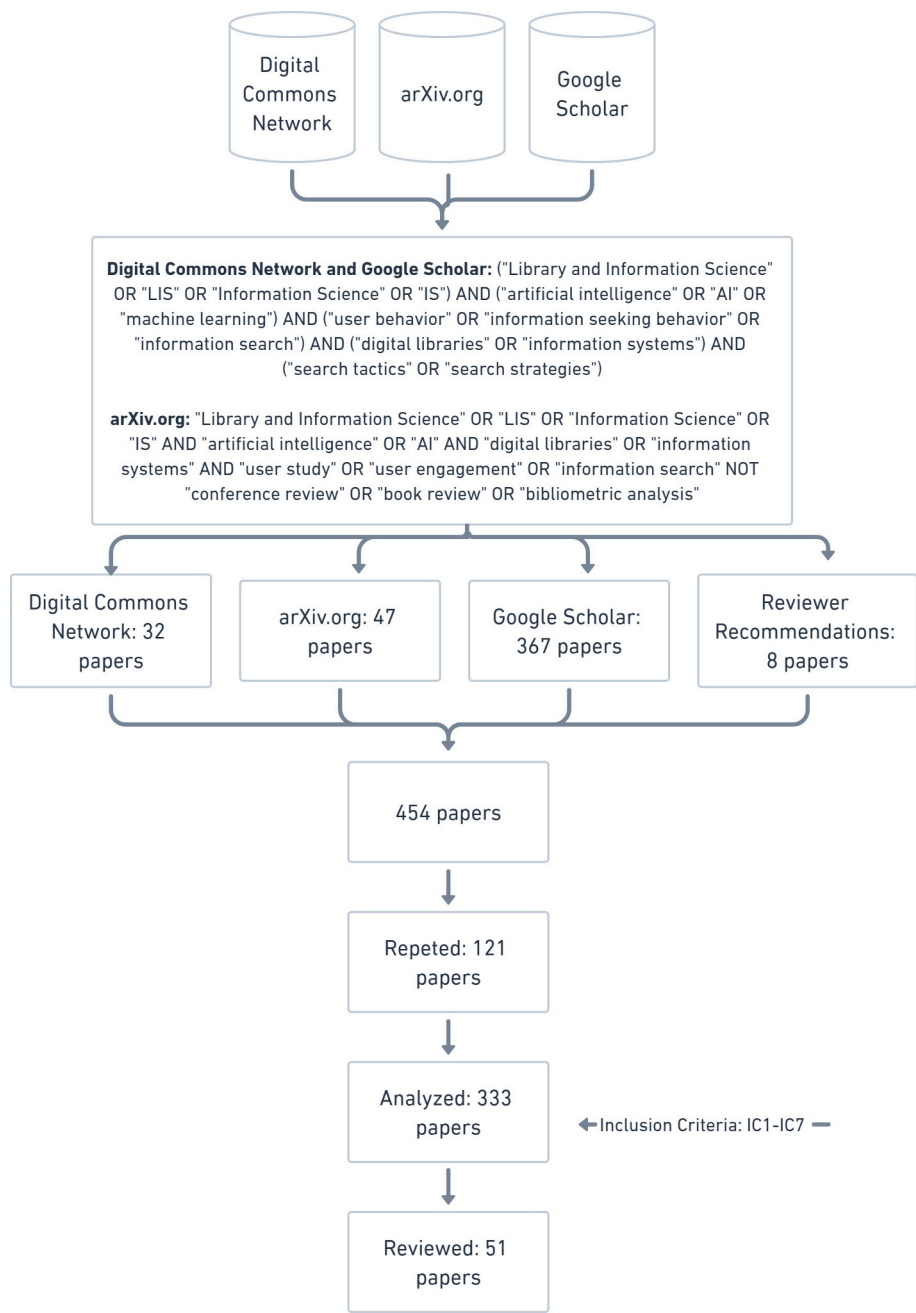


Fig. 1. Literature review process.

for inclusion were required to meet at least one of the criteria outlined in Appendix A. However, the researchers gave priority to studies that addressed multiple criteria.

Following this step, the researchers chose a total of 51 papers to be reviewed, approximately 15 % of the set. The researchers grouped the works based on three major issues: heuristics definition (including categories, tactics, and strategies); methods to gather data (e.g., how the heuristics were recorded) and to operationalize the search (e.g., the way the heuristics were applied); and decision processes (e.g., evaluation criteria and decision patterns). Fig. 2 presents an overview of each grouping. The review is presented in Appendix E. The first group shows the works concerning the heuristics mapped; the second, the works regarding methods; and the third, the ones involving decision process.

Even though the majority of search tactics were created for database searching (Bates, 1979b), they are largely technology-independent and have been adapted throughout the years to Web searching (Bates, 1987;

Smith, 2012) and to describe psychological approaches to the search process (Bates, 1979a). Eighty-seven tactics were identified at this stage, based on 29 papers, which are indicated with check marks in Fig. 2.

4.5. Analysis method

4.5.1. Stratified model to group the tactics

A model of information-seeking behavior identified the target tactics and grouped them. Several models can be identified from the literature. Still, they have in common and usually display one or more of the following premises: they have to be able to translate informational problems into statements (i.e., query statements) (Taylor, 1968); relevance feedback, a mechanism for providing information to a machine about the status of search results, is critically dependent on user interaction (Rocchio, 1971) (Belkin et al., 1995) and system implementations (Wilson et al., 2009); the already cited role of knowledge deprivation as



Fig. 2. Grouping overview.

a catalyst for information seeking (i.e., the ASK situation) (Belkin et al., 1982). More recently, the understanding that the system and the user have equal constraints on the success of a search (Saracevic, 1997) could also be considered a premise.

The user interaction side presented in Saracevic's (1997) work involved levels of cognition, representing their interpretation; attitudes, which represented their motivation; and circumstances (i.e., the situational level), which represented their informational requirements. The researchers decided to analyze the original 87 tactics based on the stratified perspective. By exposing the different layers of user interaction, each tactic was individually checked regarding the situations of the ESTI method (He et al., 2016) and its role in user-system interaction.

The mapping of tactics to specific interaction levels was based on the core function and purpose of each tactic, following the overarching structure of the three interaction levels (Appendix C): Attitude (focused on user motivation), Cognition (focused on the interpretation and decision-making process), and Circumstance (focused on the

information requirements and contextual elements of the search).

In an LLM environment, for example, cognitive strategies may involve interpreting conversational AI responses, while attitudinal strategies may focus on user motivation to refine or repeat queries. Circumstantial strategies would capture real-time adjustments based on specific content needs. The decision-making process for mapping the tactics involved the following considerations:

- **Purpose and Function:** Each tactic was evaluated based on its intended use. Tactics that were geared toward modifying user behavior or motivation (e.g., aiding, furthering, reducing, or rejecting search elements) were mapped to the Attitude level. Tactics related to interpreting and making sense of information (e.g., decision-making, interpretation, or review) were placed in the Cognition level. Finally, tactics focused on managing the search process and specific information requirements (e.g., monitoring, accessing, or browsing information) were placed in the Circumstance

level. For tactics such as Starting and Plagiarism Detection, the mapping was informed by their role in either initiating the search or ensuring ethical content usage.

- **Sublevel Mapping:** The sublevels within each interaction level further refined the categorization. For example, in the Attitude level, tactics like Backlink and Parallel were mapped to “Further the search” because they aimed to expand the search by identifying new avenues of exploration. In the Cognition level, tactics like Consult and Crosscheck were placed under “Decision making” as they directly impacted the user’s decision-making process based on the retrieved information.

Then, the researchers associated the interaction levels with the traditional categories of monitoring, file structure, search formulation, and term. In the next step, the researchers assembled the tactics following the rules presented in Appendix B, which reduced their number by compiling their intention into one indicator representing the tactic’s reasoning. Finally, the researchers reviewed the categories to check if their traditional meaning still held when considering the differences when translated from query statements to prompt statements. When not holding, the researchers created new categories to provide a suitable grouping.

4.5.2. Intercoder reliability

The researchers performed a content analysis to assess the repurposed tactics and strategies. The chosen approach was intercoder reliability, where different annotators evaluate the content and measure their agreement regarding its accuracy and consistency (Krippendorff, 2011). According to the literature, a minimum of two separate annotators was required (O’Connor & Joffe, 2020). Krippendorff (2004) stipulated two specific prerequisites: first, the annotators must have familiarity with the phenomena being examined (p.128), and second, they must maintain independence, implying no prior involvement in the research (p.121). Typically, the data sample for analysis comprises around 10–25 % of the total dataset (O’Connor & Joffe, 2020).

Three annotators participated in the annotation process and were assigned, respectively, to different data samples. The annotators held degrees in Computer Science with specializations in AI and Information Systems. Samples were randomly chosen and consisted of 15 % of the data each (45 % of the tactics were assessed). They were provided with a spreadsheet containing the tactic descriptions, category names, descriptions and codes, annotation columns, and an additional column for comments. Their task consisted of two levels of evaluation. First, they evaluated the definition of each search tactic, determining whether they agreed with it (coded as 0 for no or 1 for yes). Second, they categorized each tactic based on the predefined groupings, associating the appropriate category code with each tactic. The researchers re-coded the annotators’ responses as 0 (disagreement) or 1 (agreement) based on the original tactic groupings.

During the annotation process, some disagreements arose among the annotators, particularly concerning the categorization of certain tactics. For example, there were occasional differences in interpreting whether a tactic was better suited to one category over another. These disagreements typically involved subtle distinctions between related categories, such as determining whether a tactic belonged under “prompt formulation” or “prompting granularity”.

To address these discrepancies, the researchers facilitated a round of discussion with the annotators. The purpose was to resolve the misunderstandings and reach a consensus on ambiguous cases. Once the discussion concluded, annotators reevaluated the tactics in question based on the agreed interpretations. After the annotation process, the researchers constructed a reliability data matrix to tabulate the coincidences within the units, as described by Krippendorff (2004, p. 223–227) and Krippendorff (2011), to compute Krippendorff’s alpha (α).

According to O’Connor and Joffe (2020), values less than 0 indicated

no reliability, values between 0 and 0.20 indicated slight agreement, 0.21 to 0.40 indicated fair agreement, 0.41 to 0.60 moderate agreement, 0.61 to 0.80 substantial agreement, and values between 0.81 and 1 indicated nearly perfect agreement.

The first annotator (a1) provided an alpha of 71.43 % ($\alpha = 0.7143$). The second annotator (a2) also provided a 71.43 % agreement ($\alpha = 0.7143$), while the third annotator (a3) provided a result of 66.67 % ($\alpha = 0.6667$). The joint average result, calculated as $(\alpha a1 + \alpha a2 + \alpha a3)/3$, was 69.84 % ($\alpha = 0.6984$), indicating a substantial agreement between the annotators.

5. Findings

As a result, 45 tactics were mapped and grouped into 7 categories: tracking (7 tactics); information input (5 tactics); information output (3 tactics); information retrieval (5 tactics); prompt formulation (7 tactics); prompting granularity (12 tactics); and evaluation (6 tactics).

Tracking maintained the traditional meaning of the monitoring tactics used to keep users aware of the search’s progress; **information input** dealt with the tactics applied to provide the AI with enough information to begin working on its responses or further them; **information output** concerned the tactics used to improve or adapt the AI’s output; **information retrieval** grouped the tactics applied to stimulate the AI to browse for data; **prompt formulation** regarded the tactics used to aid specific prompt commands; **prompting granularity** concerned the actions taken to review the commands given to the AI; and **evaluation** grouped the tactics applied to assess the AI’s responses. Appendix D displays the categories, their associated tactics and the process involved.

6. Discussion

6.1. (RQ1): How can search tactics and strategies be repurposed to perform a prompt process with the aid of a tool such as ChatGPT?

In this study, the researchers applied the concepts of “topic”, “aspect”, “concept”, and “term” consistently to guide the repurposing of search tactics. Appendix G provides a more detailed explanation for each of these definitions. LLMs such as ChatGPT require a different approach to search formulation compared to traditional Boolean-based methods, primarily due to their conversational nature and the need for real-time responses. The researchers identified the repurposed tactics by considering how LLMs handle queries in natural language processing environments. The researchers redefined tactics such as “Parallel” or “Reduce” based on how prompt structure, conversational flow, and iterative improvement influence LLM responses.

Some tactics could not be repurposed. In this case, the researchers replaced them to adjust to the platform. Note that the tactics are not about content, but directives to ChatGPT. Regarding search strategies, the repurposing exclusively occurred within the metacognitive domain, which concerns skills of higher-order and content-related reflective activities during an online search (Tsai et al., 2012). The domain comprises three strategy aspects: purposeful thinking; selection of the main ideas; and evaluation.

Many of the strategies of purposeful thinking and the selection of main ideas, such as “narrowing down the searching field”, “searching information from different sources”, “typing specific terms about the search task”, and “keeping an account of the information relevant to the search task found on a website”, can be related to the first six categories of search tactics. A thorough list of these online searching strategies can be found in (Tsai & Tsai, 2003), (Reisoğlu et al., 2019), and (Tibau et al., 2023). The researchers would like to concentrate on strategies regarding the evaluation aspect, which would be important to help the user deal with generative AI’s weaknesses regarding logical flow, inaccuracy (rather in terms of both factuality and truth), bias, and lack of critical judgment and originality.

The evaluation aspect concerns the skills to “evaluate the relationships between the retrieved pieces of information”, “compare and contrast pieces of information retrieved from different websites”, “determine whether a piece of information from a website is worth referencing”, and “assess how to combine and present the data gathered”. These skills grounded the adaptation of a seventh search tactic’s category, first presented by [Smith \(2012\)](#), appropriately called “evaluate”.

6.2. (RQ2): What specific search tactics and strategies could be repurposed for this type of prompting?

Through the analysis, the researchers identified and adapted 45 distinct tactics specifically designed for use with prompting tools like ChatGPT. These adaptations were guided by the principles of prompt engineering, a practice focused on enhancing information retrieval through structured queries and optimized output control ([Liu & Chilton, 2022](#)). To provide clarity and structure, the researchers organized these tactics into categories: tracking, information input and output, information retrieval, prompt formulation and granularity, and evaluation. These categories are summarized in [Tables 1 to 7](#), with more detailed descriptions provided in Appendix F.

One of the key contributions of this study was the repurposing of traditional search tactics into strategies suited for the prompt-response cycle inherent to LLMs. For example, the researchers repurposed the traditional “Check” tactic to allow users to review whether the AI’s response corresponds with the original prompt, thereby improving control over the search process. Similarly, the “Correct” tactic encourages users to fact-check responses during or after interaction, thus ensuring accuracy. Unlike traditional search engines, LLMs require a fluid, conversational approach to information retrieval, which was referred in this study as a chat-and-search process. This shift

Table 1
Tactics for the tracking category.

Tactics	Description	Reasoning	Usage
Starting	Focuses on identifying key elements and sources to start the search.	Initiates search direction effectively for a structured approach.	Ask ChatGPT for key concepts or references to begin exploring a new research topic.
Check	Reviews the prompt against ChatGPT’s response for correspondence.	Validates the accuracy and relevance of the AI’s response.	Practice double-checking by comparing the response to online content.
Weight	Assesses the AI’s response for diversity and bias.	Highlights the importance of encountering diverse viewpoints.	Consider diversity and bias in the AI’s response.
Pattern	Examines ChatGPT’s answers for identifiable patterns to optimize prompts.	Identifies response patterns to improve prompt effectiveness.	Rewrite prompts to avoid verbosity and repetition.
Correct	Focuses on recognizing and correcting factual errors in responses.	Enhances the accuracy of information by proactive and post-interaction steps.	Fact-check and correct errors during and after the chat.
Record	Keeps track of the chat for context and productivity.	Maintains chat context and aids in tracking decisions and information.	Use labeling, summarizing, and formatting to organize chat content.
Monitoring	Continuously tracks updates and developments on the search topic.	Keeps information current and aligned with evolving needs.	Request regular updates from ChatGPT on fast-evolving fields or ongoing search queries.

Table 2
Tactics for the Information Input category.

Tactics	Description	Reasoning	Usage
Chained	Focuses on refining ChatGPT’s output through a series of building prompts.	Breaks complex prompts into simpler ones for more personalized answers.	Start with a structure prompt and progressively fill in details.
Roleplay	Defines a role for ChatGPT to simulate specific personas or situations.	Varies by aim, such as creativity or storytelling.	Use a structured prompt to define ChatGPT’s role and instructions.
Survey	Reviews AI’s recommendations at decision points during a search.	Ensures accuracy and relevance of AI responses.	Ask for suggestions on prompts and searching techniques.
URL	Uses URLs as input for contextual information or examples.	Provides context and stimulates interactive responses.	Input a URL and ask ChatGPT to base its response on it.
Backlink	Applies a pearl-growing technique for deeper exploration of a topic.	Encourages comprehensive and insightful responses.	Ask for more information based on an initial piece of relevant data.

Table 3
Tactics for the Information Output category.

Tactics	Description	Reasoning	Usage
Output	Directs how ChatGPT should present responses.	Ensures responses meet specific needs and avoids ambiguity.	Specify audience level, style, and format (e.g., as a blog entry).
Modify	Adapts text to a different tone or format.	Clarifies modification requirements and context for use.	Rewrite text for specific formats like scientific reports or articles.
Format	Instructs AI to provide responses with specific attributes.	Matches output to particular requirements or preferences.	Request responses in Markdown with structured elements like lists and tables.

Table 4
Tactics for the Information Retrieval category.

Tactics	Description	Reasoning	Usage
Social mediate	Analyzes content from social media posts.	Gains insights into public opinions and serves as inspiration.	Copy and paste social media text for analysis and advice.
Time travel	Uses cached pages or the Internet archive for insight.	Provides an educated guess on unavailable content.	Ask for analysis based on URL patterns and conventions.
Referenced	Stimulates AI to use live web information for responses.	Enables retrieval of accurate and reliable information.	Use tools like WebChatGPT plugin for web browsing.
Browsing	Encourages semi-directed exploration of related materials or topics.	Allows flexible discovery of new, unexpected, or related information.	Ask ChatGPT for suggestions on books, papers, or resources related to the current research topic.
Extracting	Systematically pulls key information from multiple sources.	Synthesizes comprehensive responses from various inputs.	Instruct ChatGPT to extract relevant data or summarize key points from several articles or sources.

necessitated adapting existing tactics to accommodate the iterative and dialogue-based nature of prompting. For example, as mentioned in subsection 6.1, the researchers redefined tactics like “Parallel” and “Reduce” based on how LLMs handle queries in natural language processing environments. The researchers adapted “Parallel” to encourage

Table 5
Tactics for the prompt formulation category.

Tactics	Description	Reasoning	Usage
Teleport	Makes AI generate structure for further prompts.	Allows AI to ask itself necessary questions.	Ask ChatGPT to write numbered questions for reference.
Exhaust	Provides detailed content in initial prompt.	More detailed prompts lead to more detailed AI responses.	Add detailed information to enrich the prompt.
Reduce	Reduces detailed information in prompt.	Clarifies focus and allows AI creative freedom.	Remove excess details to simplify the prompt.
Parallel	Broadens prompt with parallel topics.	Encourages exploration of related concepts.	Broaden tasks to include related topics for wider responses.
Pinpoint	Narrows prompt to specific topics.	Reduces ambiguity for more relevant responses.	Focus on precise aspects for targeted information.
Block	Excludes certain terms or topics.	Prevents sensitive or inappropriate content.	Instruct AI to avoid specific sensitive topics.
Mini overlap	Focuses on aspects with minimal topic overlap.	Provides targeted information on specific aspects.	Use imperative verbs to narrow down the response.

Table 6
Tactics for the prompting granularity category.

Tactics	Description	Reasoning	Usage
Specialization	Specifies AI's response with specific terms.	Improves response accuracy and coherence.	Ask for detailed answers on specific topics.
Generalization	Generalizes AI's response to broader topics.	Diversifies responses and reduces repetition.	Ask to broaden the answer across general topics.
Related	Associates AI's response with related content.	Explores related subtopics and perspectives.	Provide related terms or subjects for input.
Contrary	Changes response to the logical opposite.	Checks understanding and explores biases.	Design prompts for opposite viewpoints.
Trace	Builds upon information already given.	Generates relevant and coherent prompts.	Use past responses to formulate further questions.
Self-criticism	Prompts AI to critique its answers.	Encourages critical assessment of responses.	Ask AI to evaluate and improve its responses.
Phrase	Avoids common or vague phrases.	Enhances clarity and value of the interaction.	Use distinct and precise prompts.
Scrutinize	Ensures prompt clarity.	Helps AI generate better answers.	Make each sentence in the prompt clear.
Refine	Refines response by adding context.	Improves AI's understanding and answer accuracy.	Provide context and instructions for clarity.
Add	Adds new concepts to expand AI's scope.	Stimulates more dynamic and diverse chats.	Update prompts with new information.
Repeat	Keeps AI updated with conversation context.	Maintains narrative coherence in multi-turn chats.	Include previous interactions in new prompts.
Vary	Adjusts approach to avoid AI response cutoff.	Overcomes limitations of response length.	Instruct AI to continue from where it left off.

the inclusion of related topics within the same conversational thread, broadening the scope of information retrieved. In contrast, “Reduce” focuses on simplifying the prompt, giving ChatGPT creative freedom while narrowing the query’s scope to avoid irrelevant results.

The integration of ChatGPT with Microsoft’s Bing search engine and Google AI’s Gemini represents a significant shift in the future of search processes, blending LLM capabilities with traditional search functionalities. These developments reflect the evolution of information retrieval

Table 7
Tactics for the evaluation category.

Tactics	Description	Reasoning	Usage
Context	Establishes the setting for the AI response.	Improves interactions by ensuring response relevance.	Assess credibility, bias, and identify misconceptions.
Crosscheck	Checks AI response against other information sources.	Verifies the accuracy of the provided information.	Cross-reference answers with reliable sources.
Validation	Validates the AI response for accuracy.	Prevents the spread of misinformation.	Use tools like plagiarism detectors and search engines.
Interpretation	Understands how AI generated its output.	Enhances trust and ensures response accuracy.	Gradually change input to impact the AI response.
Mediate	Maintains awareness of biases in AI response.	Enables critical evaluation of provided information.	Look for biases in sensitive topics.
Plagiarism Detection	Ensures AI-generated content does not overlap with existing material.	Safeguards originality and ethical content usage.	Ask ChatGPT to check if generated text overlaps with known sources and ensure proper acknowledgment.

from static Boolean searches to dynamic, conversational interfaces that enhance the user experience by offering real-time, iterative responses. Such innovations could dramatically shift the foundational paradigms of information science toward an amalgamated, intelligent, and user-focused approach to information retrieval (Dinneen & Bubinger, 2021).

However, despite these innovations, challenges persist. As Formanek (2024) noted, inconsistencies in generated responses and difficulty maintaining contextual accuracy can arise when blending LLMs with traditional search engines. The repurposed tactics in this study, such as “Crosscheck” and “Validation”, offer users methods to address these challenges by providing structured strategies for verifying the reliability of AI-generated content. For example, the “Crosscheck” tactic encourages users to validate ChatGPT’s responses against external sources, while the “Validation” tactic focuses on ensuring the accuracy and reliability of AI-generated outputs through third-party verification tools.

6.3. Limitations and future work

This study provided a comprehensive review of search tactics and their repurposing for use in LLM-driven environments. However, a few limitations warrant discussion, along with the actions taken to mitigate them. Initially, the query used for the literature review was restricted to terms related to LIS and digital libraries, which may have limited the scope of retrieved papers. Although this scope was later expanded to include broader terms, such as Information Science and information search, some relevant studies might have been excluded by the original query. To address this, the researchers expanded the literature review by adding approximately 30 more papers and adjusting the query terms. This approach mitigated the initial query’s limitations and provided a more comprehensive overview of relevant studies.

The criteria for selecting papers were designed to balance specificity and breadth. However, there is potential for subjectivity in applying these criteria, particularly in determining whether a paper met one or more of the conditions outlined in Appendix A. To minimize this limitation, the researchers included all papers that met at least one of the criteria, prioritizing those that satisfied multiple criteria.

While the creation of LLM-specific search tactics represents a key contribution of this study, further research is needed to refine these tactics and ensure their applicability across various LLM platforms.

Future studies could focus on testing these tactics in multiple environments to ensure their generalizability and practical utility.

Additionally, as part of future work, the researchers plan to conduct a search-task experiment using the think-aloud method. This approach will allow researchers to observe how users apply the identified tactics in real-time, providing valuable insights into their practical utility and impact on search behaviors and outcomes. This experimental validation will contribute to refining the tactics for broader adoption in LLM-driven search environments.

7. Conclusion

Repurposing search tactics for Generative AI tools illuminates a groundbreaking path for enhancing Searching as Learning processes. By meticulously adapting established search tactics for the AI age, this work navigates the novel challenges posed by conversational agents in information retrieval and enriches the cognitive landscape of users engaging with these technologies. The introduction of 45 specifically tailored tactics heralds a significant step forward in leveraging AI's pattern recognition and novel data generation capabilities, transforming ChatGPT into a dynamic educational ally. This approach empowers users to surmount the inherent limitations of LLMs, fostering a more efficient, insightful, and critically engaged learning experience.

The implications of this study extend far beyond the immediate improvements in user interaction with AI tools; it proposes a fundamentally enhanced model of digital learning and information discovery, the chat-and-search process. Through the lens of these repurposed tactics, AI becomes not just a tool for information retrieval but a tutor that guides, challenges, and expands the user's knowledge base. However, the journey does not end here. Further research is essential to fully unlock the potential of LLMs in transforming the search into a profound learning experience, ensuring that these tools can meet the demands of diverse learning and search contexts.

CRedit authorship contribution statement

Marcelo Tibau: Writing – review & editing, Writing – original draft, Investigation, Conceptualization. **Sean Wolfgang Matsui Siqueira:** Supervision. **Bernardo Pereira Nunes:** Supervision.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.lisr.2024.101331>.

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